1.7.1 From multi-tiered to microservices architectures and beyond

IT infrastructures have always affected the way software applications are architected and designed. In the beginning, monolithic applications were deployed on huge mainframes as single components. When the internet and PCs became popular, we started designing applications according to the client/server paradigm. A multi-tiered architecture, relying on that paradigm, was widely used for desktop and web applications, decomposing the code into presentation, business, and data layers.

With the increase in application complexity and the need for agility, new ways of further decomposing the code have been explored, and a new architectural style entered the stage: microservices. In the last few years, this architectural style has become more and more popular, and many companies decided to refactor their applications according to this new style. Microservices are usually compared to monolithic applications, as shown in figure 1.13.